Proposition 1. *Let $X \in \mathbb{R}^{n\times d}$ be an input and $f_L(X) = S \circ \cdots \circ S(X)$ be a Transformer, where each layer $S(X) = S(X; W)$ corresponds to the same weight matrices W_K, W_Q, and $W_V^l = I$. Then, $f_L(X)$ converges to some matrix $f^*(X)$ of rank ≤ 1.*

Proof. We can write $f(X) = S(X; W)^k X$, and since $S(X; W)$ is a matrix whose rows are non-negative and sum to 1, its largest eigenvalue is 1 and all of the other eigenvalues lie in $(-1, 1)$. Thus, we can write $S(X; W) = PDP^{-1}$, where D is a diagonal matrix whose diagonal is the eigenvalues of $S(X; W)$ and P is an orthogonal matrix. Using this, we can write $S(X; W)^k = PD^kP^{-1}$, where D^k converges to a matrix whose diagonal is $(1, 0, \ldots, 0)$ and $S(X; W)^k$ converges to a matrix of rank ≤ 1. Thus, $f(X)$ converges to a matrix $f^*(X)$ of rank ≤ 1. □

Proposition 2. *Let $X \in \mathbb{R}^{n\times d}$ be an input and $f_L(X) = S_L \circ \cdots \circ S_1(X)$ be a Transformer, where each layer $S_l(X) = S(X; W^l)$ corresponds to randomly initialized weight matrices W_K^l, W_Q^l, and $W_V^l = I$. Assume that $f_L(X)$ converges almost surely to some random variable $f^*(x)$ with respect to the selection of the weights as a function of $L \to \infty$, and assume $f^*(X) \perp S(f^*(X); W) \cdot f^*(X)$ for W independent of $f^*(X)$. Then,* $\text{rank}(\mathbb{E}[f^*(X)]) \leq 1$.

Proof. Due to the convergence of $f_L(X)$ to the random variable $V_1 = f^*(X)$, we can write $V_1 \sim S(V_1; W^1) \cdot V_1$ with respect to the randomness of $f^*(X)$ and the parameters W. In particular, we can write $V_1 \sim V_k$ where $V_k = S(V_{k-1}; W^{k-1}) \cdot V_{k-1}$ with $W^1, \ldots, W^k$ random i.i.d. weights. In particular, $\mathbb{E}[V_1] = \mathbb{E}[V_k] = \mathbb{E}[S(V_{k-1}; W^{k-1}) \cdot \ldots \cdot S(V_1; W^1) \cdot V_1]$. Since $f^*(X) \perp S(f^*(X); W) \cdot f^*(X)$, we obtain that $V_{i+1} \perp V_i$ for all $i \in [k]$ and in particular, $S(V_i; W^i) \perp S(V_{i-1}; W^{i-1})$. Since the sequence $V_1, \ldots, V_k$ is Markovian, we have:

$$\begin{aligned}\mathbb{E}[V_k] = \mathbb{E}[V_1] &= \mathbb{E}[S(V_{k-1}; W^{k-1})] \cdot \ldots \cdot \mathbb{E}[S(V_2; W^2)] \cdot \mathbb{E}[S(V_1; W^1) \cdot V_1] \\ &= \mathbb{E}[S(V_{k-1}; W^{k-1})] \cdot \ldots \cdot \mathbb{E}[S(V_2; W^2)] \cdot \mathbb{E}[V_2] \\ &= \mathbb{E}[S(V_1; W^1)]^{k-2}\mathbb{E}[V_1]\end{aligned} \tag{2}$$

We note that for any W^1 and V_1, $S(V_1; W^1)$ is a matrix whose rows are non-negative and sum to 1, its largest eigenvalue is 1 and all of the other eigenvalues lie in $(-1, 1)$. Similarly to the proof of Prop. 1 we obtain that $\mathbb{E}[S(V_1; W^1)]^{k-2}$ converges to a matrix of rank ≤ 1, and therefore, $\mathbb{E}[S(V_1; W^1)]^{k-2}\mathbb{E}[V_1]$ also converges to a matrix of rank ≤ 1. We note that this sequence is actually constant and equal to $\mathbb{E}[V_1] = \mathbb{E}[f^*(X)]$ and therefore, $\mathbb{E}[f^*(X)]$ is a matrix of rank ≤ 1. □

When the attention weights become uniform or nearly uniform, the model may lose the ability to distinguish between important and unimportant features in the image, resulting in a loss of detail and specificity in the image representation. This can lead to less accurate predictions and less informative visualizations, such as CAM maps.

Therefore, addressing the oversmoothing problem in transformers involves preventing the attention weights from converging toward uniformity. We address the issue of oversmoothing in Transformers by forcing the attention weights to sum to 0 instead of 1.

3.2 Centered Self-Attention Layers

In the previous section, we showed that at initialization, deep Transformers suffer from oversmoothing because all of the attention matrices' eigenvalues lie in $(-1, 1)$ except for the largest eigenvalue, which is 1. In order to address the oversmoothing problem, we can offset the sum of each row by some constant $\alpha \in \mathbb{R}$ to ensure that the eigenvalues of the matrix lie in a larger interval. One way to ensure this is to offset the attention rows to sum to $1 + \gamma$ instead of 1, by the following modification:

$$S_\gamma(X; W) = \left(\text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) + \gamma\frac{\mathbb{1}\mathbb{1}^\top}{n}\right)V = \text{softmax}\left(\frac{XW_Q(XW_K)^\top}{\sqrt{d}} + \gamma\frac{\mathbb{1}\mathbb{1}^\top}{n}\right)XW_V$$

Specifically, we focus on the case where $\gamma = -1$, in which the softmax activations are adjusted to zero. We note that potentially one could obtain the same effect with different values for the offset. In order to better understand how the offset affects the degree of oversmoothing, we conducted several simulations in Section 4.1.

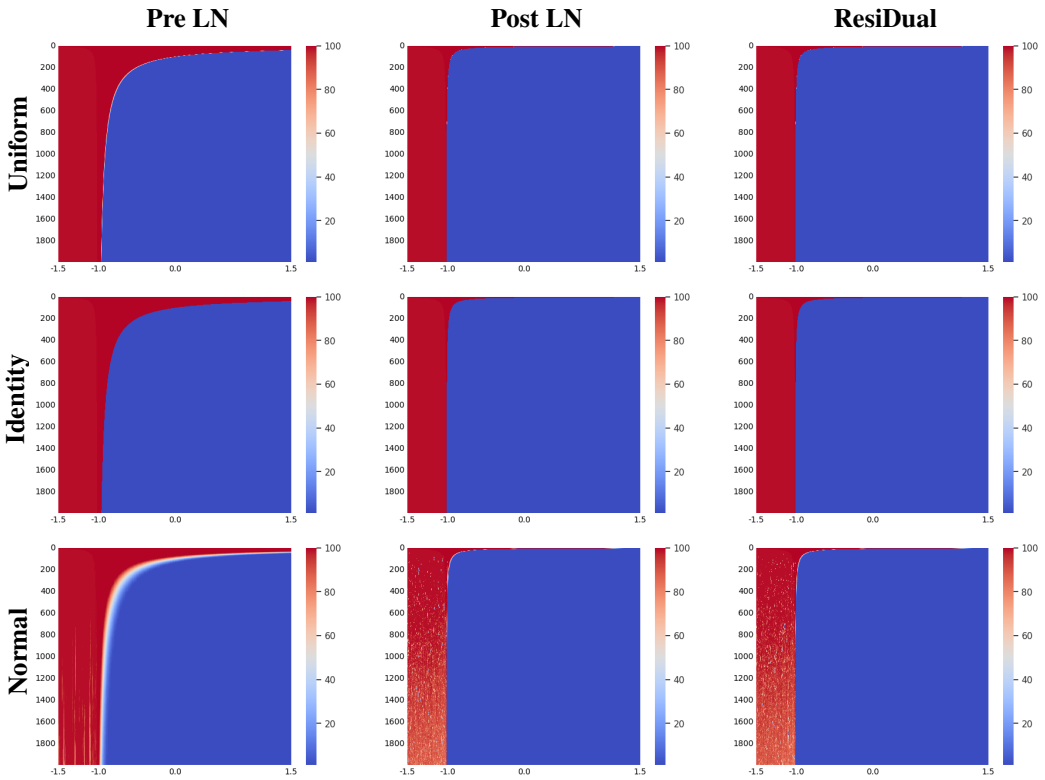

Figure 1: **Oversmoothing in Transformers.** The graph illustrates the relationship between the rank of the output and two parameters: the offset parameter γ (x-axis) and the depth of the network (y-axis). Each graph represents a specific type of Transformer architecture with different block types (Pre-LN, Post-LN, or ResiDual) and weight initialization. It is observed that the rank approaches 1 as the depth increases to infinity exclusively when $\gamma > -1$.

3.3 Application to Weakly Supervised Semantic Segmentation

The ViT model [9] involves the initial splitting of an image into patches to generate patch tokens. Subsequently, these tokens are concatenated with a learnable class token and input into the Transformer encoder, resulting in the final patch and class tokens. The multi-head self-attention is a crucial aspect of the Transformer block, facilitating global feature interaction.

The oversmoothing problem has been identified to adversely affect the output patch tokens, rendering them overly uniform after multiple Transformer blocks [51, 52, 41]. This issue severely impacts the CAMs of Transformer. Notwithstanding, the early layers of the model can still preserve semantic diversity. Consequently, following previous work [41] we have introduced an auxiliary classifier in an intermediate layer of the transformer model, alongside the application of the proposed correction term for the self-attention layers, to counteract the oversmoothing issue. The resulting loss function can be written as $\mathcal{L} = \mathcal{L}_{cls} + \mathcal{L}^m_{cls}$, where $\mathcal{L}_{cls}$ and $\mathcal{L}^m_{cls}$ are Multilabel soft-margin losses of the last layer and the intermediate layer classifiers.

In order to achieve single-stage weakly supervised semantic segmentation, we extract the class activation maps from the last classification layer. These maps are later refined using a pixel-adaptive refinement module (PAR) [40] to generate final refined pseudo labels, using the concept of pixel-adaptive convolution [1, 46] to improve the accuracy of pseudo labels by efficiently incorporating RGB and position information of nearby pixels.

The refined pseudo labels are then utilized to supervise the segmentation decoder, which is trained via a Cross-Entropy loss ($\mathcal{L}_{seg}$). Additionally, a regularization loss term [47] ($\mathcal{L}_{reg}$) is used, following prior research on single-stage weakly supervised semantic segmentation [41, 34, 39]. The regularization

Table 1: **VOC Semantic Segmentation Results**. $Sup.$ denotes the supervision type. $\mathcal{I}$: Image-level labels; $\mathcal{S}$: Saliency maps. $Net.$ denotes the backbone network (for single-stage methods) and the semantic segmentation network (for multi-stage methods). † denotes using ImageNet-21k [36] pretrained parameters.

Method	$Sup.$	$Net.$	val	test	Method	$Sup.$	$Net.$	val	test
Multi-stage methods.					***Multi-stage methods (continued)***.				
RIB [24]	$\mathcal{I}+\mathcal{S}$	DL-V2	70.2	70.0	MCTformer [59]	$\mathcal{I}$	WR38	71.9	71.6
EPS [26]	$\mathcal{I}+\mathcal{S}$	DL-V2	71.0	71.8	ESOL [28]	$\mathcal{I}$	DL-V2	69.9	69.3
L2G [19]	$\mathcal{I}+\mathcal{S}$	DL-V2	72.1	71.7	***Single-stage methods***.				
RCA [63]	$\mathcal{I}+\mathcal{S}$	DL-V2	72.2	72.8	RRM [60]	$\mathcal{I}$	WR38	62.6	62.9
Du [10]	$\mathcal{I}+\mathcal{S}$	DL-V2	**72.6**	**73.6**	1Stage [1]	$\mathcal{I}$	WR38	62.7	64.3
RIB [24]	$\mathcal{I}$	DL-V2	68.3	68.6	AFA [39]	$\mathcal{I}$	MiT-B1	66.0	66.3
ReCAM [41]	$\mathcal{I}$	DL-V2	68.4	68.2	SLRNet [34]	$\mathcal{I}$	WR38	67.2	67.6
VWL [39]	$\mathcal{I}$	DL-V2	69.2	69.2	ToCo [41]	$\mathcal{I}$	ViT-B†	71.1	72.2
W-OoD [25]	$\mathcal{I}$	WR38	70.7	70.1	**Ours**	$\mathcal{I}$	Deit-S†	**72.5**	**72.9**

Table 2: Evaluation and comparison of the semantic segmentation results in mIoU on the val set. † denotes using ImageNet-21k [37] pretrained weights.

	bkg	***aero***	***bicycle***	***bird***	***boat***	***bottle***	***bus***	***car***	***cat***	***chair***	***cow***	***table***	***dog***	***horse***	***motor***	***person***	***plant***	***sheep***	***sofa***	***train***	***tv***	***mIoU***
1Stage [1]	88.7	70.4	35.1	75.7	51.9	65.8	71.9	64.2	81.1	30.8	73.3	28.1	81.6	69.1	62.6	74.8	48.6	71.0	40.1	68.5	64.3	62.7
AFA [39]	89.9	79.5	31.2	**80.7**	67.2	61.9	81.4	65.4	82.3	28.7	83.4	41.6	82.2	75.9	70.2	69.4	53.0	85.9	44.1	64.2	50.9	66.0
ToCo [41]	89.9	**81.8**	35.4	68.1	62.0	76.6	83.6	80.4	87.7	24.5	**88.1**	54.9	87.0	84.0	76.0	68.2	65.6	85.8	42.4	57.7	65.6	69.8
ToCo† [41]	**91.1**	80.6	48.7	68.6	45.4	**79.6**	**87.4**	**83.3**	89.9	35.8	84.7	60.5	83.7	83.2	**76.8**	**83.0**	56.6	87.9	43.5	60.5	63.1	71.1
Ours†	89.3	81.1	**61.2**	78.6	**73.0**	75.4	83.1	71.0	**90.4**	**55.2**	84.7	**76.9**	**87.9**	**85.8**	72.5	70.5	**65.9**	**88.5**	**60.4**	**68.9**	**69.9**	**72.5**

Table 3: **Evaluation of pseudo labels**. † denotes using ImageNet-21k [37] pretrained parameters.

Method	1Stage	AA&LR	SLRNet	AFA	ViT-PCM	ViT-PCM + CRF	ToCo	**Ours**
Backbone	WR38	WR38	WR38	MiT-B1	ViT-B†	ViT-B†	ViT-B†	Deit-S†
mIoU	66.9	68.2	67.1	68.7	67.7	71.4	73.6	**74.1**

Table 4: **Ablation Study. M**: CAM from the final layer; $\mathbf{M}^m$: auxiliary CAM from the intermediate layer; $Seg.$: semantic segmentation results.

			w/o centering term			with centering term		
Layers	$\mathcal{L}_{seg}$	$\mathcal{L}_{reg}$	$\mathbf{M}$	$\mathbf{M}^m$	$Seg.$	$\mathbf{M}$	$\mathbf{M}^m$	$Seg.$
$\mathcal{L}_{cls}$			28.1	–	–	56.1	–	–
$\mathcal{L}_{cls}+\mathcal{L}^m_{cls}$			26.3	57.8	–	57.2	64.6	–
$\mathcal{L}_{cls}+\mathcal{L}^m_{cls}$	✓		66.4	67.8	66.7	70.1	65.3	70.4
$\mathcal{L}_{cls}+\mathcal{L}^m_{cls}$	✓	✓	68.0	66.3	68.9	71.4	63.5	71.8

loss term serves to enforce spatial consistency in the predicted segmentation masks, reducing the risk of segmented regions with irregular shapes or disconnected segments.

The details of the training pipeline are given in the supplementary appendix. At test time, in order to further improve the segmentation performance, the proposed method employs multi-scale testing and dense CRF processing, which are standard techniques in semantic segmentation. While this description may seem a little elaborate, the method is entirely composed of conventional components that are used throughout the relevant literature. As our ablation study shows, without using our term, this pipeline by itself is not very effective.